

Karthik Manu

Metallurgical Researcher



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Work Experience

Researcher, Swerim AB

10/2022 – present | Luleå, Sweden

Led research on valorisation of steel industry residues, biocarbon applications, and metal recovery from slags and battery black mass, while managing key industrial partnerships and aligning their needs with research priorities at Swerim

Master Thesis Student, Swerim AB

2022 | Luleå, Sweden

Studied mechanical and reduction extent of biocarbon agglomerates, incorporating MODDE software

Shift Manager, Velove

2021 | Stockholm, Sweden

Led and managed riders during shifts, ensuring smooth operations and effective coordination

Education

Masters Degree: Engineering Materials Science, KTH Royal Institute of Technology

2020 – 2022 | Sweden

Bachelors Degree: Mechanical Engineering, Amrita Vishwa Vidyapeetham

2017 – 2021 | India

Software

FactSage, MS Office, MODDE, MATLAB

Summary

Metallurgist by passion, materials scientist by education, and a researcher by curiosity, I focus on unlocking value from waste streams through metal recovery, slag valorisation, and sustainable agglomeration using bio-based solutions, supported by strong project management skills.

Honors

Jernkontoret Scholarships

Awarded multiple scholarships and travel grants by Jernkontoret for competence development

EU RFCS project

Authored and contributed to two European-funded projects, coordinating consortia of over eight partners

Publications

Maximizing the Recycling of Iron Ore Pellets Fines Using Innovative Organic Binders

2023

Journal of Materials MDPI

Titanium in Cast Cu – Sn Alloys – A review

2021

Journal of Materials MDPI

Solutionising temperature influence on the morphological and mechanical characteristics of Al-Si-Mg-Ni hypoeutectic alloys

2021

Journal of Institution of Engineers (India) Series-D

Morphological and mechanical behaviour of Cu-Sn alloys - A Review

2020

Metals And Materials International (MAMI)

References

Dr. Elsayed Mousa, Senior Specialist,
Metallurgy Department, Swerim AB
Elsayed.Mousa@swerim.se

Dr. Björn Glaser, Professor Process Metallurgy,
Royal Institute of Technology
bjoerng@kth.se

Languages

English, Swedish, Hindi

Interests

Scuba diving, Pencil Sketching, Photography